

DS-973 — Data Science & Machine Learning Essentials

COMPREHENSIVE REFERENCE MATERIALS & CHEAT SHEET

Course Code: DS-973	Category: Computer Science & Tech
Academic Term: 2026 Academic Year	Format: E-Learning / Self-Paced
Prerequisites: Basic Problem Solving	Estimated Hours: 40 Hours Content

1. Core Concepts & Theoretical Framework

This reference document provides quick access to core syntax, APIs, design principles, and best practices for **Data Science & Machine Learning Essentials**. Keep this sheet accessible as you work through lab exercises and projects.

Fundamental Rules & Design Conventions:

- 1. Clean & Readable Code:** Prioritize clear variable naming, modular design, and concise function responsibilities.
- 2. Error Handling & Robustness:** Always validate inputs, handle edge cases, and catch runtime exceptions gracefully.
- 3. Efficiency & Complexity:** Pay attention to Time Complexity $O(N)$ and Space Complexity constraints.
- 4. Documentation:** Maintain inline docstrings, type annotations, and README instructions.

2. Quick Reference Syntax & Code Templates

```
# Example 1: Standard Initialization & Workflow
def process_course_data(records: list) -> dict:
    results = {"total": len(records), "valid": 0, "processed": []}
    for record in records:
        if record.get("status") == "active":
            results["valid"] += 1
        results["processed"].append(record["id"])
    return results

// Example 2: Asynchronous Execution & API Handling
async function fetchCourseDetails(courseId) {
    try {
        const response = await fetch(`/api/courses/${courseId}`);
        if (!response.ok) throw new Error(`HTTP Error: ${response.status}`);
        const data = await response.json();
        return data.course;
    } catch (error) {
        console.error("Failed to load course details:", error);
        return null;
    }
}
```

3. Recommended Reading & Official Documentation

- Official Documentation:** Official language specifications, API docs, and standard library guides.
- Recommended Textbook:** *Clean Code: A Handbook of Agile Software Craftsmanship*
- Design Patterns:** *Elements of Reusable Object-Oriented Software*
- Community Resources:** Developer Hubs, StackOverflow, and GitHub repositories.